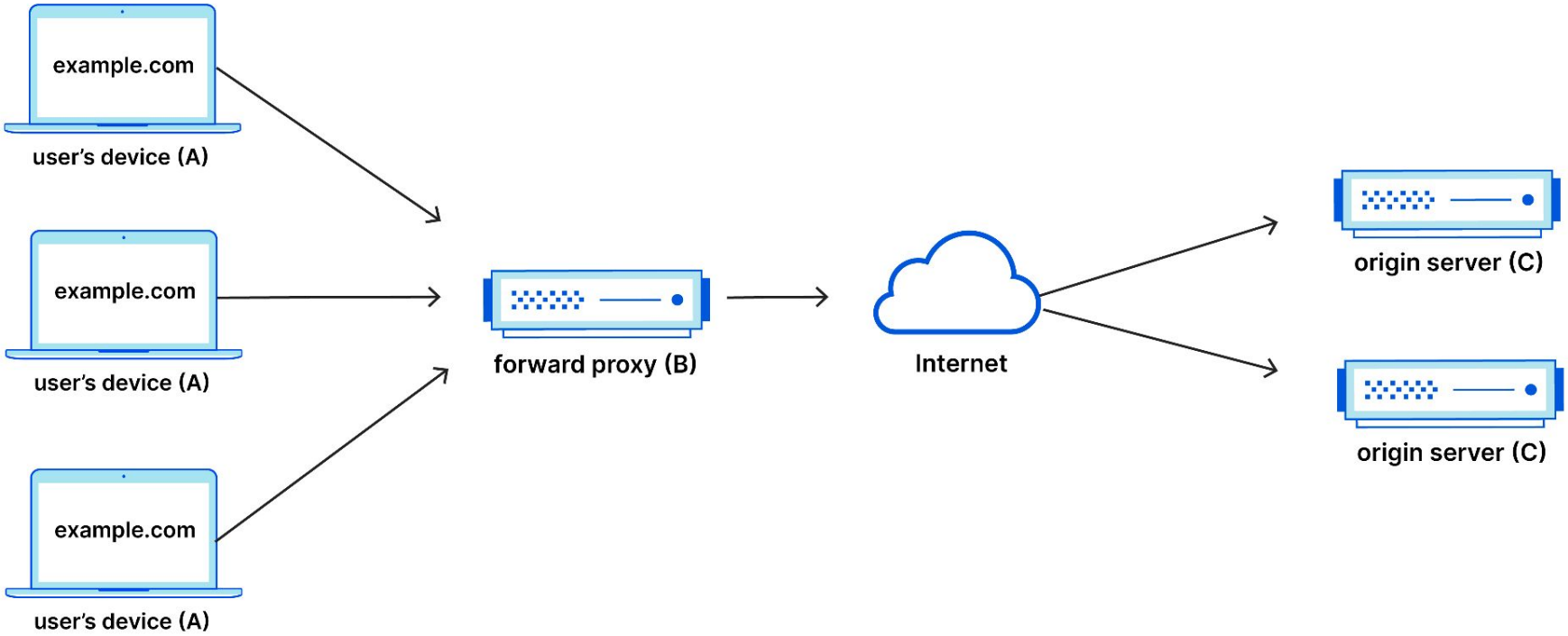
Proxies
{forward, reverse} proxy

Spring 2025

Proxy

- Proxy
  - the authority to represent someone else, especially in voting.
- Forward proxy, Proxy server, Web proxy
  - All the same
- An application-layer intermediary that intercepts, processes, and forwards client requests while optionally modifying traffic, enforcing policies, or optimizing delivery.
  - Sits between the **client** and **server**
  - Main purpose is to process requests from clients and forward them to appropriate server, and relay the response back.
- Used for various purposes
  - Enhancing security
  - Improving performance
  - Providing anonymity
- Instead of connecting to the server directly, we connect **through** a proxy.

Forward Proxy Flow



Forward proxy

- A server that acts as an intermediary between the client and the wider internet.
  - Clients send request to proxy server, which forwards them to the internet and returns the response to the client.
- Used to **control** and **filter** outgoing traffic.
  - e.g. restricting access to certain websites
- Can use **caching** to improve performance by caching frequently accessed content
- Can provide anonymity by hiding the client's address from the destination
  - Because you are connecting **through** the proxy now.

Use cases

- Schools usually use them for **content filtering**
- ISPs or some corporate networks use them for **caching**
- Most of us use it for **anonymity** or to overcome censors.
- It can also be used for security, where it can filter malicious content before it reaches to the user.

Problems

- They may not support HTTPS, which can cause potential security issues.
- If they are used for caching, caching strategies are important.
  - Data can become outdated.
- Does not guarantee anonymity.
  - Many proxies leak information.
  - We need to use special proxies for that (high-anonymity proxies)

Reverse proxy

- It sits between client and server again, but this time it sits in front of servers.
  - It forwards the client requests to webservers.
  - Increase security, performance and reliability.
- FP sits in front of clients, RP sits in front of servers.
- When a client sends a request to a server, RP catches it on the **network edge**
- Load balancer is an example.

Good for

- Good for security
  - The website never reveals their address. All our requests are going to the RP.
  - Therefore the attackers can only attack the RP.
- Good for performance
  - It is possible that a website is distributed on several servers around the world
  - RP can send the request where it is closest to the client.
- Caching
  - Can cache content, which means if it has the information it may just return it back to the user.
  - In caching, caching strategies are always important.

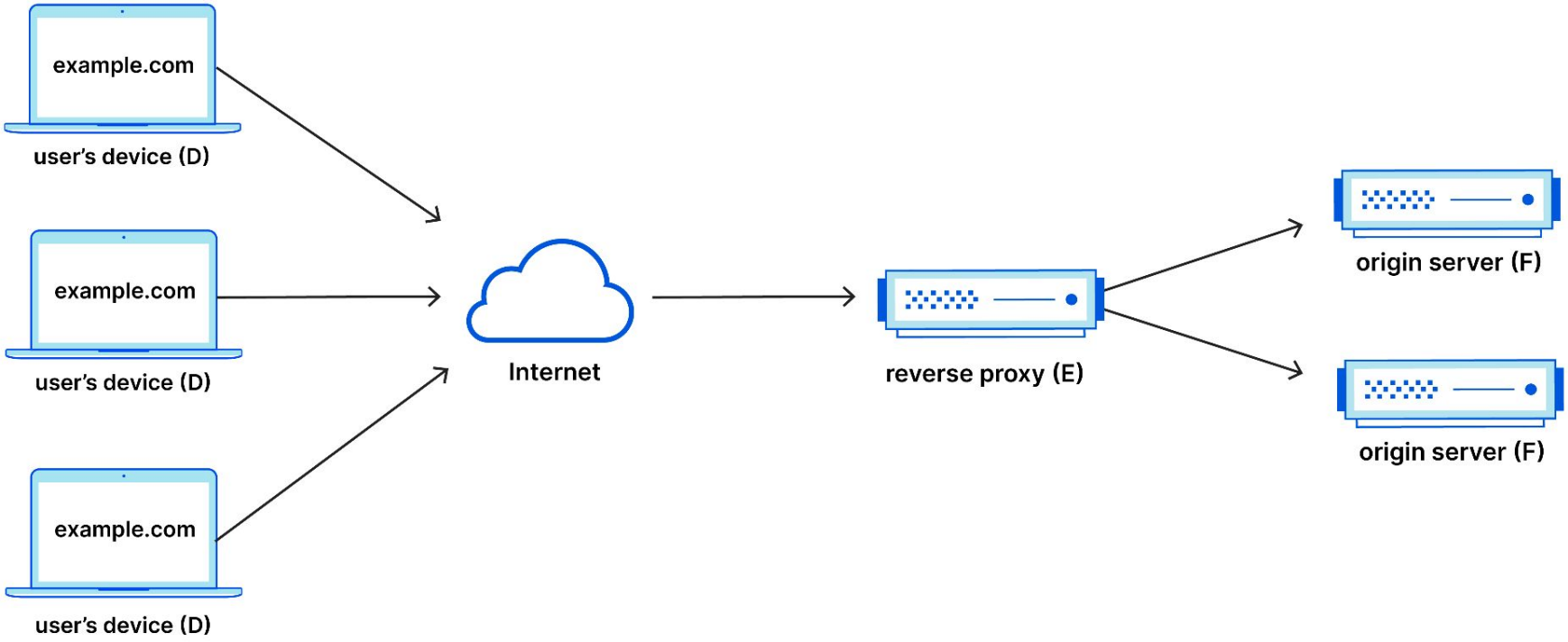
Good for

- Even if there are multiple backend servers, they appear as one!
- Can handle TLS encryption and decryption
  - Reduces load on backend
- Load balancing
  - Handles and distributes load across multiple servers

Disadvantages

- Single point of failure
  - If the RP goes down, clients cannot access backend servers.
- Setting them up can be complex.
- Slight overhead
  - Now we have a middleware and therefore some latency.
- Misconfiguration can create security problems.

Reverse Proxy Flow



Feature	Forward Proxy	Reverse Proxy
Who it protects	The Client (User)	The Server
Visibility	Client knows it's there (usually)	Client thinks it's the real server
Key Use Case	Anonymity, bypassing filters	Load balancing, security
Location	Internal network or local	Edge of the server's network

VPN: Virtual Private Network

- VPN in Opera is not real.
  - It is a proxy.
  - A HTTPS proxy, encrypts the traffic between the *browser* and *proxy*
- VPN:
  - Operates at the operating system level
  - Every traffic leaving your device is encrypted
  - OpenVPN, Wireguard etc. at *network level*
  - For all installed apps